

Project Implementation Summary

Course: CPE 508 — Computer System Project Management

Institution: Obafemi Awolowo University (OAU)

Framework: React + TypeScript + Vite + Tailwind CSS

Backend Services: Firebase Auth & Cloud Firestore

1. Executive Overview

PastQ-Hub is a comprehensive, production-grade web application engineered to bridge the gap between conventional study methods and interactive self-evaluation for Computer Engineering students at OAU. Traditional study resources (paper past papers, unorganized PDFs) lack structured answers and interactive study loops. PastQ-Hub solves this by introducing a centralized, verified digital repository of exam questions combined with an adaptive quiz engine, role-based resource uploading, and collaborative forums.

2. Technology Stack & Architecture

The system utilizes a modern serverless architecture tailored for high performance and low latency on mobile and desktop web browsers:

- **Frontend Development:** Built on **React 18** with **TypeScript**, compiled via **Vite** for rapid hot-reloading and build optimization. Icons are powered by **Lucide React**.
- **UI & Styling:** Responsive components styled with utility classes via **Tailwind CSS** augmented by custom animations (e.g. chimes, check indicators) and standard design variables.
- **Authentication:** Integrated with **Firebase Authentication** supporting both standard secure email/password registration and Google OAuth sign-in popups.
- **Database (Cloud Firestore):** Utilizes **Google Cloud Firestore** as a NoSQL document database, indexing collections for fast query execution on courses, questions, user bookmarks, and discussion threads.

3. Core Features & Capabilities

Feature Component	Implementation Details & Technical Capabilities
Authentication & Auth Guards	Provides secure registration, logins, and logouts. Enforces strict role-based authorization (Student, Lecturer, Administrator) to control access to upload and review dashboard views.
Interactive Quiz Engine	Allows users to configure and take timed tests on specific past questions (Objective, Theory, Fill-in-the-Blank, and Calculation formats). Shows scores, detailed solutions, and step-by-step calculations upon submission.
Analytics & Exam Trends	Tracks the historical frequency of topics (e.g. Earned Value Management, Critical Path Method) across multiple exam years. Labels them dynamically based on popularity (Moderate, High, Very High).
Discussion Forum	Enables peer-to-peer and student-lecturer collaboration. Supports thread creation, content formatting, reply counts, and real-time updates of discussions related to complex academic questions.
Past Question Repository	Houses approved past question files, allowing students to filter questions by Course Code, Topic, Year, Semester, and Question Type. Features a Bookmark system for saving hard questions for later review.
Upload & Admin Dashboard	Authorized lecturers and admins can submit past question files in JSON/CSV formats. The Admin Dashboard implements a file validation/approval panel where submissions can be approved or discarded before publishing.

4. Project Management & Evaluation Metrics

As a project developed within the scope of **CPE 508 (Computer System Project Management)**, the application embeds active formulas and calculators for core project monitoring techniques. Specifically, it features an interactive ****Earned Value Management (EVM)**** module that explains:

- **Cost Performance Index (CPI)** and **Schedule Performance Index (SPI)** metrics.
- **Cost Variance (CV)** and **Schedule Variance (SV)** calculations.
- Real-world interpretations (e.g. $CPI > 1$ indicates under budget; $SPI < 1$ indicates behind schedule).

By implementing these calculations as interactive quiz questions with complete step-by-step solution breakdowns, the system acts as a practical learning tool, helping students master EVM calculations through hands-on practice.

5. Future Roadmap

- ****PDF Export Module:**** Allow students to download full solved past question papers as offline PDFs.
- ****AI Tutor Integration:**** An embedded chatbot model to explain complex calculation questions dynamically based on student inputs.
- ****Automated Grading:**** Using LLMs to evaluate student answers in theory-based questions and assign marks.